

Contoso Electronics

Equipment Reference & Maintenance Guide

CO2 Laser Cutter

Contoso PrecisionCut LC-150

Asset Identification

Field	Detail
Asset ID	CE-LAS-3300
Category	Fabrication
Manufacturer	Contoso Fabrication Systems
Location	Enclosure Fabrication Shop - Bay 2
Installation Date	2022-08-22
Warranty Expiry	2025-08-22
Support Contact	fab.support@contoso.com ext. 4380

Overview

The Contoso PrecisionCut LC-150 is a CO2 laser cutting and engraving system used for fabricating enclosures, panels, gaskets, and acrylic fixtures. It combines a sealed CO2 laser source with a motion gantry and integrated fume extraction for clean, precise cuts.

Technical Specifications

Parameter	Value
Laser Type	Sealed CO2, 150 W
Work Area	1300 x 900 mm
Cutting Speed	Up to 1000 mm/s
Positioning Accuracy	±0.02 mm
Cooling	Closed-loop water chiller
Assist Gas	Compressed air / nitrogen
Power	230 VAC, 50/60 Hz, 3.5 kW

Preventive Maintenance Schedule

Interval	Task
Daily	Clean lens and mirror; check chiller water level and temp.
Weekly	Inspect belts and rails; lubricate linear guides.
Monthly	Verify beam alignment; clean exhaust ducting.

Annual	Replace chiller coolant; service laser tube if power drops.
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Safety Guidelines

- Class 4 laser - never bypass safety interlocks or open lid during operation.
- Always run fume extraction; laser-cut materials release hazardous fumes.
- Keep a fire extinguisher nearby; never leave the machine unattended while cutting.
- Only cut approved materials - never cut PVC or chlorinated plastics.

Common Issues & Troubleshooting

Symptom	Recommended Action
Incomplete cuts	Clean lens, check focus height, verify laser power and speed.
Chiller alarm	Check water level and ambient temperature; clear airflow.
Reduced engraving quality	Realign mirrors; inspect lens for burns.
Axis not homing	Check limit switches and gantry obstructions; inspect drive belt.

Contoso Electronics internal maintenance documentation. For service requests, contact fab.support@contoso.com | ext. 4380 or submit a work order through the maintenance portal.